The resistivity of materials is

i) inversely proportional to the number density (n) of the electrons

ii) inversely proportional to the average time between the collisions (τ).

In metals, if the temperature increases, the average time between the collision (τ) decreases and n is independent of temperature. In semiconductors when temperature increases, n increases and τ decreases, but increase in n is dominant than decreasing τ, so that overall resistivity decreases.



The resistance of certain materials become zero below certain temperature T_c. This temperature is known as critical temperature or transition temperature. The materials which exhibit this property are known as superconductors. This phenomenon was first observed by Kammerlingh Onnes in 1911. He found that mercury exhibits superconductor behaviour at 4.2 K. Since R = 0, current once induced in a superconductor persists without any potential difference.

EXAMPLE 2.13

If the resistance of coil is 3 Ω at 20 ^{0}C and α = 0.004/^{0}C then determine its resistance at 100 ^{0}C.

Solution

R_0 = 3 Ω, T = 100^{0}C, T_0 = 20^{0}C

α = 0.004/^{0}C, R_T = ?

$R_T = R_0(1 + \alpha(T\text{-}T_0))$

$R_{100} = 3(1 + 0.004 \times 80)$

$R_{100} = 3(1 + 0.32)$

$R_{100} = 3(1.32)$

$R_{100} = 3.96\ \Omega$

EXAMPLE 2.14

Resistance of a material at 10^{0}C and 40^{0}C are 45 Ω and 85 Ω respectively. Find its temperature co-efficient of resistance.

Solution

$T_0 = 10\ ^0C$, $T = 40\ ^0C$, $R_0 = 45\ \Omega$, $R = 85\ \Omega$

$$\alpha = \frac{1}{R_\circ}\frac{\Delta R}{\Delta T}$$

$$\alpha = \frac{1}{45}\left(\frac{85-45}{40-10}\right) = \frac{1}{45}\left(\frac{40}{30}\right)$$

$$\alpha = 0.0296\, per^\circ C$$

2.3 ENERGY AND POWER IN ELECTRICAL CIRCUITS

When a battery is connected between the ends of a conductor, a current is established. The battery is transporting energy to the device which is connected in the circuit. Consider a circuit in which a battery of voltage V is connected to the resistor as shown in Figure 2.15.

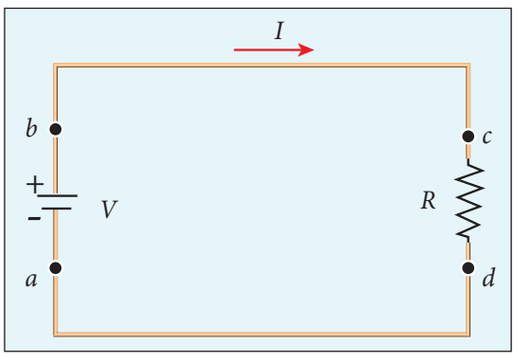


Figure 2.15 Energy given by the battery

Assume that a positive charge of dQ moves from point a to b through the battery and moves from point c to d through the resistor and back to point a. When the charge

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